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Effects of an Acetone Extract of *Boswellia carterii* Birdw. (Burseraceae) Gum Resin on Rats with Persistent Inflammation

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ABSTRACT

Objective: *Ruxiang*, or *Gummi olibanum,* an herbal medicine derived from the gum resin of *Boswellia carterii* Birdw. (BC) of the family Burseraceae, has been used traditionally in China to alleviate pain and reduce inflammation. The present study is an investigation of the effects of a BC extract on persistent hyperalgesia and edema in rats with peripheral inflammation.

Design: In this randomized, blinded study, the antihyperalgesic and antiedema effects of 3 dosages of BC were compared to a vehicle control. Inflammation was induced in rats by injecting complete Freund's adjuvant (CFA) into one hind paw. A single oral dose of the BC extract was administered daily for 7 days, beginning one day before CFA. Hyperalgesia was assessed using a paw withdrawal latency (PWL) test pre-CFA and 2 hours, 5 hours, 1 day, and 5 days post-CFA. Edema was determined by measuring paw thickness at the same time points. Spinal Fos protein expression was analyzed 2 hours post-CFA. Adverse effects of the extract were monitored by observing the animals closely for unusual behavioral changes.

Results: Compared to control, a dosage of 0.45 g/kg BC significantly lengthened PWL and reduced paw edema on day 5 post-CFA. At 0.90 g/kg, BC significantly lengthened PWL at 5 hours, 1 day, and 5 days, and reduced paw edema at 2 hours, 5 hours, 1 day, and 5 days. This dosage also significantly suppressed spinal Fos expression in the medial half of laminae I–II. At 1.80 g/kg, BC significantly lengthened PWL and reduced paw edema at all time points.

No noticeable adverse effects were observed in animals given the lower dosages of BC, but adverse effects in some animals were observed at 1.80 g/kg per day. In the acute toxicity study, the maximal single dose of 2.50 g/kg produced no adverse effects in the treated rats during the 14 days of observation.

Conclusions: The data suggest that BC produces significant antihyperalgesia and anti-inflammation effects and that the antihyperalgesia may be mediated by suppressed inflammation-induced Fos expression in the spinal dorsal horn neurons.

INTRODUCTION

Chronic inflammatory diseases such as arthritis and low back pain affect a large population of patients in the United States (American College of Rheumatology, 1996; Palm et al., 2001; Pincus and Callahan, 1993; Silverstein, 1998). Although conventional medicine has made considerable progress in the treatment of inflammatory disease, nonsteroidal anti-inflammatory drugs (NSAIDs) and recently developed COX-2 inhibitors are often associated with significant adverse effects such as gastrointestinal disturbances (Buffum and Buffum, 2000; Pomp, 2002; Scheiman, 2001; Stiel, 2000). Complementary and alternative medicine (CAM), which has increasingly gained in popularity in recent years (Barnes et al., 2004; Eisenberg et al., 1998), may provide additional treatment options. *Ruxiang*, or *Gummi olibanum,* the dried gum resin obtained from *Boswellia carterii* Birdw. (BC), is one of 43 species of the genus *Boswellia*

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of the family Burseraceae (International Plant Name Index Query, 2004). BC has long been employed as an herbal medicine in Traditional Chinese Medicine (TCM); its earliest recorded use is documented in *Miscellaneous Records of a Famous Physician* (*Min Yi Bie Lu*, written by Tao Hong-Jing between 480 and 498 AD).

In TCM theory, the function of BC is to invigorate the blood and promote the movement of *qi*. It has been employed traditionally for over 2000 years to treat inflammatory diseases and pain conditions such as tissue injury, trauma, or arthritis-like disorders (Bensky and Gamble, 1993; Zhu, 1998). A related species, *Boswellia serrata* Roxb. ex Colebr. (Indian frankincense tree), is the source of an extensively studied dried gum resin (BS), used in Ayurvedic medicine for the management of arthritis (Singh and Atal, 1986). An extract of BS has reportedly shown noticeable anti-inflammatory, antiarthritic (Singh and Atal, 1986; Sharma et al., 1988; Sharma et al., 1989), and possibly immunomodulatory activities (Sharma et al., 1996).

BC is widely used in herbal formulas (Qu et al., 1997; Wang et al., 1994; Xu et al., 1997; Zhou and Fang, 2003), and commercial herbal products containing it are easily available in the United States as dietary supplements for patients with arthritis or other inflammation- or pain-related disorders (McGuffin et al., 1997). Nevertheless, there is little reported scientific investigation of its antihyperalgesic and anti-inflammatory properties.

In this study, our purpose was to investigate the effect of BC and its mechanisms of action in a complete Freund's adjuvant (CFA) animal model of persistent, inflammatory pain. The recently developed CFA-induced persistent pain and inflammation animal model is ideal for assessing the separate effects of antihyperalgesia and anti-inflammation in the same animal (Lao et al., 2001). CFA-induced persistent inflammation mimics subacute rheumatoid arthritis (RA), which is a disease characterized by excessive synovial-based immunologic activity (Firestein, 1997). Fos protein expression has been used successfully in animal models to study the transmission and/or modulation of noxious messages in the central nervous system (CNS) (Bullitt, 1990; Hunt et al., 1987; Lao et al., 2001).

Our specific aims were to determine the antihyperalgesic and antiedema effects of a standardized BC acetone extract, and the effect of BC on spinal transmission and/or modulation of noxious messages by observing Fos protein expression in the spinal cord of the CFA-inflamed rat model.

MATERIALS AND METHODS

Animal preparation

Male Sprague-Dawley (SD) rats weighing 250 to 270 g (Harlan, Indianapolis, IN) were purchased at least 1 week before the experiment. The animals were kept under controlled environmental conditions (22°C ± 0.5°C, relative humidity 40 to 60%, 12-hour alternating light-dark cycles, food and water *ad libitum*). The rats for the hyperalgesia study were gently handled for 30 minutes and acclimatized another 30 minutes in a clear plastic chamber (5" × 8" × 11") each day beginning 2 days before the behavioral baseline test. Animal protocols were approved by the Institutional Animal Care and Use Committee of the University of Maryland, Baltimore. The ethical guidelines of the International Association for the Study of Pain for the treatment of animals were followed in these experiments (Zimmermann, 1983).

Inflammation and hyperalgesia were induced by injecting CFA (suspended in an 1:1 oil/saline emulsion, 0.08 mL, 0.04 mg heat-killed *Mycobacterium tuberculosis*) subcutaneously into the plantar surface of one hind paw of the rat with a 25-gauge hypodermic needle (Lao et al., 2001). The inflammation, manifesting as redness, edema, and hyper-responsiveness to noxious stimuli, was limited to the injected paw. Inflammation appeared shortly after the injection, peaked between 6 hours and 24 hours, and lasted about 2 weeks. CFA-inflamed rats showed routine grooming behavior and levels of activity, and the effect of hyperalgesia on their normal behavior seemed minimal.

Herbal preparation

Unprocessed BC (gum resin), purchased at Tong Ren Tang, an herbal store in Beijing, China, was identified according to the latest pharmacopoeia of the People's Republic of China (2000). The raw herb was ground into powder (~80 mesh) and processed at room temperature with 70% aqueous acetone to completely extract both nonpolar and polar components. To minimize the possible breakdown of thermolabile compounds that may be present in extracts, evaporation temperatures were kept below 55°C. This initial crude extract was concentrated under reduced pressure and the dried residue was coded, weighed, and divided into two portions: one for bioassay studies, the other to be stored in a deep freezer for future reference. The final extract was 40% the weight of the raw herb.

The quality of the BC extract was monitored by thin layer chromatography (TLC) and high performance liquid chromatography (HPLC) to ensure lot-to-lot consistency. The independent mass spectroscopy core laboratory at the university also tested the extract with HPLC and mass spectroscopy (MS) prior to the experiments. Consistency was confirmed by HPLC using bioactive markers and by MS using a signal to noise ratio of at least 4:1 over background. The HPLC trace of the extract is showed in Figure 1.

Herbal treatment procedures

In the behavioral study, rats were randomly divided into 4 groups (n = 8 per group) for treatment with the BC ex-

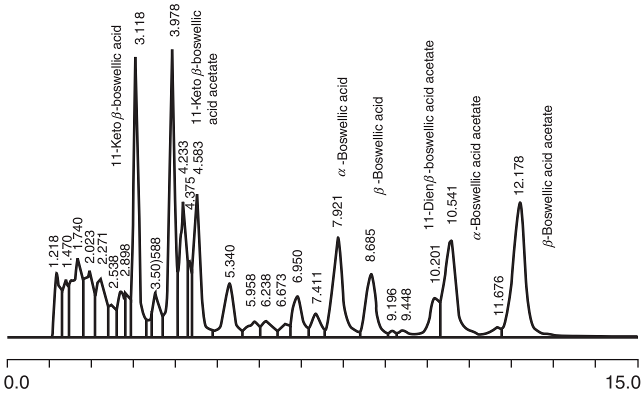

FIG. 1. High performance liquid chromatography (HPLC) trace of the initial *Boswellia carterii* Birdw. (BC) acetone extract. The HPLC conditions were: YMC C-18 analytical column (4.6 mm × 15 cm); 100% methanol at 1.0 mL/minute; UV monitor at 210 nm. Retention times of purified boswellic acids are delineated directly on this tracing.

tract at daily dosages of 0.45 g/kg, 0.90 g/kg, and 1.80 g/kg, and for treatment with vehicle control. Each dose of the extract was dissolved in 2 mL of sesame oil and administered intragastrically (i.g.) using a 5 mL syringe with a 4 cm long gavage needle through the mouth to the stomach. Animals received 1 dose on each of 7 days according to the following timetable: 1 day before, 1 hour before, and daily on each of the 5 days following CFA injection. Each animal in the vehicle control group received a daily dose of 2 mL sesame oil on the same schedule. The administration was performed gently and swiftly, and no analgesics were required for the procedure.

The initial dosage of 0.45 g/kg was based on body surface area and was calculated from the maximal daily human clinical dosage (15 g of the raw herb in an herbal formula) (Wang et al., 2002; Zhang, 1918) according to the equation: Human dosage × extract conversion factor (40%) × rat conversion factor (0.018) ÷ rat's weight in kg = rat dosage, in this case, $15\ g \times 0.40 \times 0.018 \div 0.25\ kg \approx 0.45\ g/kg$ daily (Yu, 2000).

For the Fos protein immunohistochemistry study, rats were randomly divided into the following groups ($n = 5$ per group): CFA plus vehicle control, CFA plus BC at 0.45 g/kg, CFA plus BC at 0.90 g/kg, and BC alone at 0.90 g/kg on naive rats. The BC extract or vehicle was given i.g. twice, 1 day and 1 hour prior to the CFA injection.

Thermal hyperalgesia test and inflammatory edema measurement

Rats were tested for hind paw thermal hyperalgesia by a method described in detail by Hargreaves et al. (1988). Hyperalgesia was determined by a decrease in paw withdrawal latency (PWL) to a noxious thermal stimulus. Briefly, rats were placed individually under an inverted clear plastic chamber (5 × 8 × 11 inches) on the elevated glass surface (30°C) of a Paw Thermal Stimulator System (UCSD, San Diego) and allowed to acclimatize for 30 minutes. A radiant heat stimulus was applied by a high intensity projector lamp bulb (CXL/CXR, 8 V, 50 W) located underneath the glass floor. The heat stimulus was directed onto the plantar surface of each hind paw. The intensity of the thermal stimulus was adjusted to 4.95 Ampere to derive an average baseline PWL of approximately 10.0 seconds in naive animals. The reaction of the animal to the thermal stimulus was simply to withdraw its paw from the thermal source when it felt pain, at which time the PWL to the nearest 0.1 second was automatically determined. A 20-second cut-off was used to prevent tissue damage. PWL of the inflamed paw was usually 4 to 6 seconds faster than that of the normal paw, indicative of a state of thermal hyperalgesia.

PWL was established by averaging the latencies of 4 tests, with a 5-minute interval between each test. PWL tests were conducted once (prior to gavage) on the day prior to CFA injection, twice (2 hours and 5 hours post-CFA) on the day of the injection, and twice (immediately prior to and 1 hour after gavage) on days 1 and 5 post-CFA. Inflammatory hind paw edema was determined by measuring paw thickness with a caliper 1 day prior to CFA injection and 2 hours, 5 hours, 1 day, and 5 days post-CFA. All behavioral measurements were performed by investigators who were blinded to treatment group assignments.

Immunohistochemistry

The immunohistochemistry study was conducted 2 hours after CFA injection. Rats were deeply anaesthetized with sodium pentobarbital (100 mg/kg, intraperitoneally) and perfused transcardially with 4% paraformaldehyde (Sigma) in 0.1 mol phosphate buffered saline (PBS) at pH 7.4. The lumbar 4–5 ($L_{4–5}$) spinal cord was removed, immersed in the same fixative for 2 hours at 4°C, and transferred to 30% sucrose (w/v) in PBS overnight for cryoprotection. Sections 30μ-thick were cut with a cryostat at -20°C. Free-floating tissue sections were rinsed in PBS with 0.75% Triton X-100 and 1% H_2O_2 for 1 hour, blocked with 3% normal serum (NS) in PBS for 30 minutes, then incubated with anti-Fos (1:20,000, Ab-5, Oncogene) antibody with 1% NS overnight at room temperature. After two washes in PBS (10 minutes each), the sections were incubated with biotinylated secondary IgG (Vector, 1:400 in PBS) for 60 minutes. Following two more 10-minute rinses in PBS, the sections were incubated in avidin and biotinylated horseradish peroxidase complex (Vector, 1:100 in PBS) for 60 minutes. After two more 10-minute washings in PBS, the tissue sections were reacted with 0.05% diaminobenzidine dihydrochloride (DAB, Sigma) in 0.1mol PBS containing 0.003% hydrogen peroxide for 6 minutes. Finally the sections were rinsed in PBS, mounted on gelatin-coated slides, dehydrated in alco-

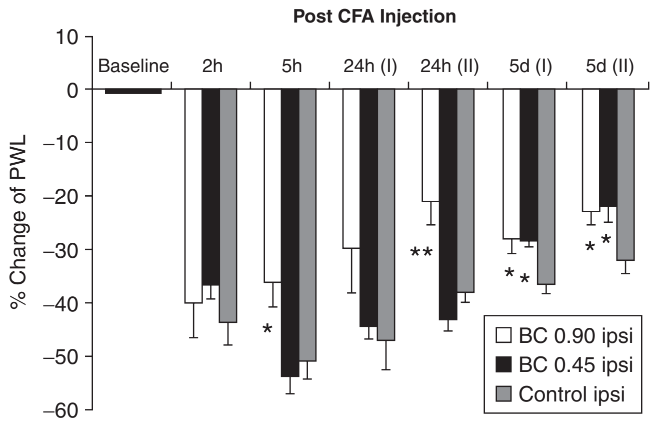

FIG. 2. Time course of the effect of *Boswellia carterii* Birdw. extract (BC) (0.45 g, 0.90 g/kg daily) on inflammatory hyperalgesia. The extract was given once a day for 7 days starting one day prior to complete Freund's adjuvant (CFA). Paw withdrawal latency (PWL) was measured immediately before (I) and 1 hour after (II) herbal administration. Values are mean ± SEM (n = 8/group). *$p < 0.05$, **$p < 0.01$ versus vehicle control.

hol, cleared in xylenes, and coverslipped with DPX (Electron Microscopy Sciences). Fos-immunoreactive neurons were identified by their densely stained nuclei. The investigator responsible for counting the Fos immunoreactive neurons was blinded to the treatment assignment of each animal. Control sections without the primary antisera produced no positive staining.

Toxicity/adverse effects assessment

Standard pharmacological categories of toxic or adverse behavioral reactions were adopted in our experiment according to the methods described by Gad (2002). Animals were closely monitored for unusual behavioral changes and for such symptoms as loss of appetite, diarrhea, weight loss, vomiting, fur discoloration, sedation, irritation, and convulsion during the 7 days of the repeated-dose experiment. To evaluate the acute toxicity of BC after a single oral dose, 20 rats were treated with 2.50 g/kg, the maximum feasible dosage that can be dissolved in 3 mL vehicle sesame oil, which was the maximal volume of sesame oil that did not cause diarrhea in the rats. Mortality, clinical signs, body weight, and behavioral activities were monitored for 14 days (Cao et al., 2001; Food and Drug Administration, 2004; Gad, 2002; Shen and Sun, 2000; Shin et al., 2003) after herbal administration.

Statistical analysis

The number of animals used for each specific test was estimated via a power analysis using either our own preliminary data or published results employing similar experimental procedures and physiologic parameters (Bausell and Li, 2002). The results are presented as mean ± standard error of the mean (SEM). In all experiments, analysis of variance (ANOVA) was used to reveal significant differences among groups. All *post hoc* comparisons were conducted using Dunnett's test; $p < 0.05$ was considered significant.

In the immunohistochemistry study, the Fos-immunoreactive neurons of 10 randomly selected L_{4-5} sections from each animal were counted on both sides of the spinal cord under a light microscope with a 20× objective lens and then averaged as the raw data for that rat. The following regions of spinal gray matter were delineated: laminae I–II medial, I–II lateral, III–IV, V–VI, VII–IX, and X.

RESULTS

Effects of BC on hyperalgesia and edema

Before BC administration, there were no significant baseline differences in PWLs ($F = 0.135$, $p = 0.87$) and paw thickness ($F = 1.0$, $p = 0.39$) among the groups. Following CFA injection, PWLs of the injected paws dramatically decreased during the first 24 hours as inflammation and hyperalgesia manifested and peaked, then gradually subsided to approach baseline values by day 5 ($F = 37.79$, $p < 0.001$) (Fig. 2).

Hyperalgesia

Post-CFA, BC at 0.45 g/kg/day significantly prolonged PWL on day 5 versus control (before BC, 7.64 ± 0.19 seconds versus 6.76 ± 0.18 in control, $p = 0.005$; after BC, 8.31 ± 0.12 versus 7.25 ± 0.20, $p = 0.001$). At 0.90 g/kg/day, BC significantly prolonged PWL at 5 hours (6.85 ± 0.44, versus 5.24 ± 0.32 in control, $p = 0.027$), 24

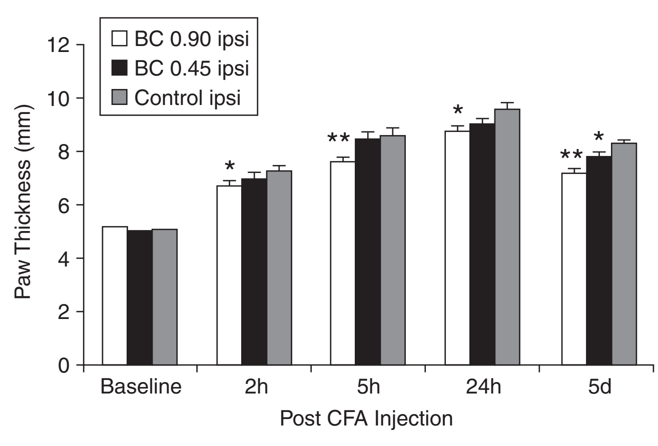

FIG. 3. Time course of the effect of *Boswellia carterii* Birdw. extract (BC) (0.45 g, 0.90 g/kg daily) on complete Freund's adjuvant (CFA)-induced paw edema. The extract was given once a day for 7 days starting one day prior to CFA. Paw thickness was measured 1 hour after herbal administration using a caliper. Data are expressed as the mean values ± SEM (n = 8/group). *$p < 0.05$, **$p < 0.01$ versus vehicle control.

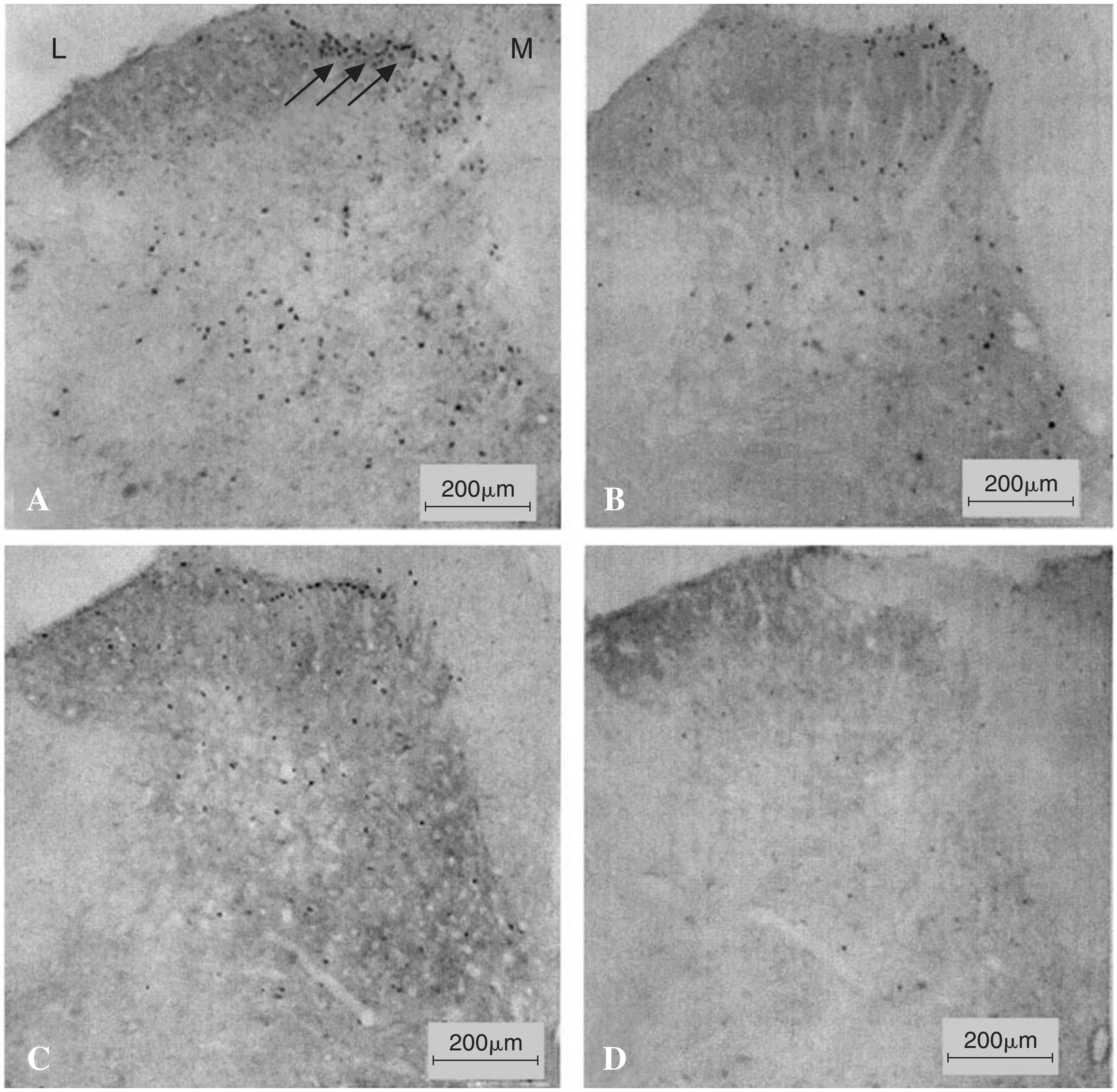

FIG. 4. Microphotographs, obtained by a Nikon microscope (E600) with a digital camera (DXM1200), showing Fos expression in dorsal horn neurons of the $L_{4–5}$ spinal cord 2 hours after complete Freund's adjuvant (CFA) injection into left hind paws of Sprague-Dawley (SD) rats. **A.** Fos was markedly induced in the dorsal horn on the side ipsilateral to the peripheral inflammation in vehicle control rats. *Boswellia carterii* Birdw. extract (BC) dosage-dependently inhibited CFA-induced Fos expression in the medial part of the spinal laminae I–II at a dosage of 0.45 g/kg is shown in **B**. at 0.90 g/kg is shown in **C**. **D**. BC alone induced little Fos in naive rats. M = medial, L = lateral; arrows indicate the medial part of superficial laminae I–II in the spinal dorsal horn, Fos-positive cells were identified as neurons containing dark stained nuclei. Original magnification, ×20.

hours (8.49 ± 0.43 versus 6.62 ± 0.40 in control, $p = 0.001$), and 5 days (before BC, 7.78 ± 0.32 versus 6.76 ± 0.17 in control, $p = 0.022$; after BC, 8.30 ± 0.25 versus 7.25 ± 0.23 in control, $p = 0.015$).

In the 1.80 g/kg/day group, BC also markedly increased PWL at all time points (data not shown); however, this experiment was terminated after 2 rats were tested, as they showed obvious adverse effects.

In the contralateral (noninflamed) hind paw, no changes from baseline PWL were observed in the vehicle control group at any time point tested ($F = 0.074$, $p = 0.93$). However, in the 0.90 g/kg/day group, PWL of the contralateral paw significantly increased after herbal administration post-CFA at 5 hours (11.72 ± 0.56 versus 10.09 ± 0.30 in control, $p = 0.028$), 24 hours (11.47 ± 0.38 versus 10.26 ± 0.33 in control, $p = 0.032$), and 5 days (11.14 ± 0.21 versus 10.46 ± 0.31 in control, $p = 0.002$), $F = 3.63$, $p < 0.001$.

Edema

Paw edema appeared shortly after the CFA injection, peaked within 24 hours, and gradually subsided by day 5.

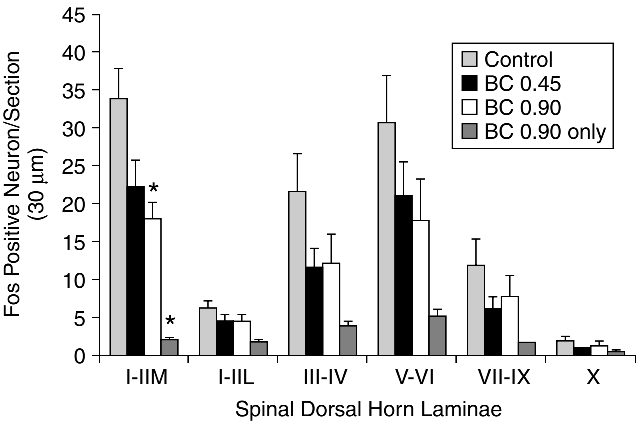

FIG. 5. Effect of *Boswellia carterii* Birdw. extract (BC) on spinal Fos expression. Ten L_{4-5} sections were randomly selected from each animal. The Fos-immunoreactive neurons were counted under a light microscope, and the values were averaged on both sides of the spinal cord ($n = 5$ per group). BC at 0.9 g/kg significantly inhibited complete Freund's adjuvant (CFA)-induced Fos in the medial part of the spinal laminae I–II versus vehicle control. BC alone induced little Fos in naive rats. F = 5.10, $p = 0.017$. *$p < 0.05$.

The extent of the edema was reduced in BC-treated groups versus control (F = 70.09, $p < 0.001$).

In the 0.45 g/kg/day BC group, paw edema was significantly less than in control on day 5 (7.84 ± 0.21 and 8.38 ± 0.11, respectively, $p = 0.028$). In the 0.90 g/kg/day group, paw edema was significantly less than in control at all time points: at 2 hours, 6.78 ± 0.17 versus 7.35 ± 0.15 in control, $p = 0.02$; at 5 hours, 7.18 ± 0.18 versus 8.67 ± 0.26 in control, $p = 0.006$; at 24 hours, 8.81 ± 0.24 versus 9.65 ± 0.21 in control, $p = 0.024$; at 5 days, 7.26 ± 0.19 versus 8.38 ± 0.11 in control, $p < 0.001$ (Fig. 3).

Similarly, in the 1.80 g/kg/day group ($n = 2$), paw edema was less at almost all time points than that of control (data not shown).

Effect of BC on spinal Fos protein expression

Fos-positive neurons were identified as neurons containing dark stained nuclei (Fig. 4). The effect of BC on spinal Fos protein expression is shown in Figure 5. Few Fos-positive neurons were seen on either side of the spinal cord in the L_{4-5} spinal cord in noninflamed naive control rats treated with BC at 0.90 g/kg (2.08 ± 0.26/section). In contrast, larger numbers of Fos-positive neurons were found on the ipsilateral side of the spinal cord in CFA-injected vehicle control rats, mainly in laminae I–II medially (33.88 ± 4.27/section) and in laminae V–VI. BC at 0.90 g/kg also significantly suppressed medial Fos protein expression in laminae I–II ipsilaterally (18.17 ± 2.26/section, $p = 0.01$). However, at the half dosage (0.45 g/kg), Fos protein expression was only slightly suppressed (22.05 ± 3.70/section) and statistical significance was not reached ($p = 0.16$). No significant effect on Fos protein expression was observed at any dosage in any other region on either side of the spinal cord ($p > 0.05$).

Adverse effects and toxicity of BC

During the 7 days of daily repeated-dose BC treatment, no noticeable abnormal animal behavioral changes or weight loss were observed in either the vehicle control or the 0.45 g/kg/day treatment groups. However, mild sedation, fur discoloration, some diarrhea, and slight weight loss were observed in some animals treated with 0.90 g/kg/day. In the two rats given 1.80 g/kg/day, obvious adverse reactions and signs were observed, including sedation, marked fur discoloration, frequent diarrhea, pale paw skin, no desire to access food or water, and marked weight loss. In the acute toxicity study, one oral i.g. administration of 2.50 g/kg BC, the maximal feasible dosage, showed no effect on animal mortality, clinical signs, or body weight in any of the 20 tested animals during the 14-day observation.

DISCUSSION

In TCM, BC is one of the standard herbs used to reduce swelling and alleviate pain (Bensky and Gamble, 1993). The present study demonstrated that an extract of BC attenuates CFA-induced hyperalgesia and edema in a dosage-dependent manner. BC at 0.45 g/kg per day significantly suppressed hyperalgesia only in the later stage (day 5) of inflammation, while the higher dosage of 0.90 g/kg per day showed significant antihyperalgesia during both early (5–24 hours) and later (day 5) stages.

The appearance of antihyperalgesic effects after 6 daily treatments of BC at 0.45 g/kg suggests that antihyperalgesia may be cumulative at this lower dosage in persistent pain conditions. It is worth noting that BC at 0.90 g/kg on day 1 post-CFA showed antihyperalgesia 1 hour after administration but not before treatment, indicative of quick action. Lengthened PWL in the paw contralateral to the CFA injection suggests that BC also increases the normal nociceptive threshold, which indicates an analgesic effect. The time course of the effect of BC on CFA-induced paw edema correlates positively to that of the hyperalgesia. We speculate that the antihyperalgesic effect of BC is likely associated with its anti-inflammatory effect, although BC may have a direct antihyperalgesic effect to some extent at the higher dosage.

CFA-induced inflammation mimics subacute rheumatoid arthritis (RA), which is a disease characterized by excessive synovial-based immunologic activity (Firestein, 1997). The anti-inflammatory effects shown in the present study suggest that BC may have immunomodulatory properties. Our HPLC analysis of the initial BC extract showed that its main chemical constituents are boswellic acids, consistent with previous reports (Badria et al., 2003; Zhou and Cui, 2002). It has been reported that boswellic acids selectively decrease the formation of leukotriene B_4, a potent chemoattractant and activator of both granulocytes and macrophages (Safayhi et al., 1992; 2000) and reduce the infiltration of leucocytes into an inflammation site (Sharma et al., 1989). The BC extract induces a dose-dependent inhibition of T-helper type 1 (Th1) cytokine production by murine splenocytes (Chevrier et al., 2002). Several *in vivo* and *in vitro* studies on the effects of boswellic acids derived from BC have shown anticancer activities and potential immunomodulatory activities (Jing and Han, 1993; Jing et al., 1999; Liu and Qi, 2000; Zhu, 1998). It is well known that granulocytes and macrophages produce pro-inflammatory cytokines such as TNF-α, IL-1, and IL-6, which play important roles in inflammation (Feldmann et al., 1996) and inflammation-induced hyperalgesia (Woolf et al., 1997). Therefore, we hypothesize that BC inhibits the infiltration of leucocytes and the production of pro-inflammatory cytokines, consequently alleviating inflammation and inflammation-induced hyperalgesia. However, the effects of BC on cytokines in the animal model need further investigation.

The mechanism of BC antihyperalgesia may be mediated by the CNS. Our data show that BC dosage-dependently suppressed ipsilateral CFA-induced Fos expression in the medial laminae I–II of the spinal cord, where most nociceptive primary afferents terminate. This suggests that BC may produce analgesic and antihyperalgesic effects by inhibiting the transmission of noxious messages at the spinal level. Evidence demonstrates that spinal Fos expression is controlled by the supraspinal descending inhibitory system (Ren and Ruda, 1996; Zhang et al., 1994). BC may activate the supraspinal descending inhibitory system, thus resulting in the inhibition of spinal Fos expression and subsequent antihyperalgesia.

Concerns over the safety and possible side effects of herbs have been raised in the medical community (Foster et al., 2000; Matthews et al., 1999; WHCCAMP, 2002). Although our acute toxicity study showed that BC had no significant adverse effects, animals given a high daily dosage of BC showed obvious adverse effects. This suggests that optimal dosages of herbal medicine are crucial in order to achieve maximum effects with the fewest adverse effects when treating patients with inflammatory pain disorders. The optimal dosage of BC determined in this study, 0.90 g/kg per day, is the equivalent of a human dosage of 30 g of raw BC, far more than the 10 to 15 g/day commonly prescribed clinically as part of an herbal formula. The smaller dosage suggests that the interactions among the various herbs in a formula may exert an additive or synergistic effect to neutralize the toxicities or side effects of the individual constituents (Bensky and Barolet, 1990). However, scientific research on the interaction among herbs in such formulas is largely lacking and needs to be pursued.

This study provides support for the clinical utility of BC as an herb that reduces swelling and alleviates pain (Bensky and Gamble, 1993). Our study provides evidence for these effects, and studies by other researchers also suggest that BC, either alone or as part of a formula, may be clinically useful in inflammatory diseases. For example, Zhen et al. (2003) reported that aqueous or ethanol extracts and raw BC significantly suppress inflammatory edema of the ear and significantly attenuate abdominal contractions induced by acetic acid in mice. Pharmacodynamic tests of Chinese herbal formulas containing BC on various inflammatory and pain animal models include studies by Qu et al. (1997), Wang et al. (1994), Xu et al. (1997), and Zhou and Fang (2003). Moreover, other Chinese herbs, similarly classified as reducing swelling and allaying pain, also show effects against inflammation and hyperalgesia. For example, Wei et al. (1999) reported that crude water extracts of the herb *Duhuo* (*Radix Angelicae pubescentis*) significantly attenuated CFA-induced edema and hyperalgesia from 24 hours to 1 week. Dai et al. (2000) reported that crude water extracts of the Chinese herbal formula *Huang-Lian-Jie-Du-Tang* produced similar results. The findings of the present study and research reported by others support the traditional use of herbs in this category and confirm the functions documented by TCM clinical practice and in the literature.

Our data may provide useful information for the clinical use of this Chinese herb in treating patients with persistent inflammatory pain and inflammatory diseases such as RA. Further exploration of the mechanisms of BC, used alone and in interaction with other herbs in a formula, is warranted.

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